Covariance Estimators and Out-of-Sample Hedging of SOFR Swap Risk

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PCA hedging and covariance estimation

- Principal component analysis (PCA) hedging neutralizes a portfolio's exposure to the few factors that drive most yield-curve movements.
- Litterman and Scheinkman (1991) found that three factors, interpreted as level, slope and curvature, explain most of the variation in US Treasury returns, so a hedge against the leading factors needs only as many swaps as factors.
- The hedge and the risk that remains after it are both computed from an estimated covariance matrix of daily rate changes.
- The sample covariance matrix weights a year of changes equally. Ledoit and Wolf (2004) shrink it toward a scaled identity matrix, and Chen et al. (2010) derive the oracle approximating shrinkage (OAS) weight for Gaussian samples.
- An exponentially weighted moving average (EWMA) instead discounts older changes, and RiskMetrics uses a decay factor of 0.94 for daily data.

Research question

Question

Does the choice of covariance estimator affect the hedging effectiveness of a SOFR swap portfolio and the accuracy of its out-of-sample risk forecasts, using daily rates from 14 swap tenors spanning 2024 to 2025?

Finding

1. No estimator consistently outperforms the equal-weighted sample in hedging effectiveness.
2. Shrinkage does not improve PCA hedges and worsens minimum-variance hedges with two swaps. EWMA improves 36 or 37 of the 79 PCA hedges with two swaps and worsens 6 to 10.
3. The estimator matters for the risk forecast. Equal weighting overstates the unhedged risk as volatility decreases in 2025, EWMA with a decay factor of 0.94 shows no significant bias, and shrinkage overstates the residual risk of hedged portfolios by a factor of 2.5 to 4.

Data

Half-year	PC1 sd (bp)	PC1 share (%)
2024 H1	21.4	95.3
2024 H2	19.5	93.0
2025 H1	20.5	93.7
2025 H2	13.7	93.2

PC1 of daily SOFR swap rate changes by half-year.

- Daily SOFR swap rates at 14 tenors from 2 to 30 years, 2 January 2024 to 31 December 2025.
- Two holidays with partial quotes, 19 February and 4 July 2024, are removed, and daily changes are taken in basis points (bp).
- The first principal component (PC1) explains 93% to 95% of the variance in every half-year.
- Its daily standard deviation falls from about 20 bp to 13.7 bp in the second half of 2025.

Hedges

$$\boldsymbol{d} = (-5.0, -2.0, -1.0, 3.5, -0.1, 1.0, 2.3, -0.8, -0.6, 2.5, 3.0, 4.0, 1.5, 5.0)^\top \times 10^3$$

- $\boldsymbol{d}$ is the dollar value of 1 bp (DV01) of the portfolio by tenor, from 2 to 30 years, with a net DV01 of \$13,300.

$$\boldsymbol{q} = \boldsymbol{d} + B\boldsymbol{h}, \qquad \pi_{t+1} = -\boldsymbol{q}^\top \Delta \boldsymbol{y}_{t+1}$$

- $k \in \{1, 2\}$ swaps at the tenors selected by $B \in \mathbb{R}^{p \times k}$, $p = 14$, hold the DV01s $\boldsymbol{h}$, and π_{t+1} is the profit and loss (P&L) of the hedged DV01 $\boldsymbol{q}$ for the rate changes $\Delta \boldsymbol{y}_{t+1}$.

$$\boldsymbol{h}_{\text{PCA}} = -(V_k^\top B)^{-1} V_k^\top \boldsymbol{d}, \qquad \boldsymbol{h}_{\text{MV}} = -(B^\top \hat{\Sigma} B)^{-1} B^\top \hat{\Sigma} \boldsymbol{d}$$

- The PCA hedge sets the exposure to the k leading eigenvectors V_k of $\hat{\Sigma}$ to zero, and the minimum-variance hedge minimizes $\boldsymbol{q}^\top \hat{\Sigma} \boldsymbol{q}$.
- The hedge sets are the 14 single tenors and the 79 of 91 pairs whose condition number $\kappa(V_2^\top B)$ is below 10 in the first estimation window.

Covariance estimators

- The equal-weighted estimator is the sample covariance matrix S of the $W = 252$ daily changes before day t.
- LW (Ledoit and Wolf, 2004) and OAS shrink it toward a scaled identity matrix,

$$\hat{\Sigma} = (1 - \delta)S + \delta\tau I, \qquad \tau = \mathrm{tr}(S)/p$$

- The EWMA estimator applies the RiskMetrics covariance recursion to all changes before day t, with the weights normalized to sum to one,

$$\hat{\Sigma} = \sum_{j \geq 1} w_j \, \Delta \boldsymbol{y}_{t-j} \Delta \boldsymbol{y}_{t-j}^{\top}, \qquad w_j \propto \lambda^{j-1}$$

- The decay factor λ is 0.94, the RiskMetrics value for daily data, or 0.98906, which gives a half-life of 63 trading days.

Evaluation

Every test day t from 3 January to 31 December 2025 is hedged with the estimates from the changes before it.

$$\mathrm{RR} = \frac{\sum_t \pi^2_{t,\mathrm{hedged}}}{\sum_t \pi^2_{t,\mathrm{unhedged}}}$$

- The residual risk RR is reported as the median over each hedge set.
- For each hedge, the squared P&L under each estimator is compared with that under the equal-weighted estimator by the Diebold and Mariano (1995) test, and the hedges significantly better or worse at the 5% level are counted.

$$z_t = \frac{\pi_t}{\sqrt{\boldsymbol{q}^\top \hat{\boldsymbol{\Sigma}} \boldsymbol{q}}}, \qquad \mathrm{H}_0 \colon \ \mathbb{E}\left[z_t^2\right] = 1$$

- The mean $\overline{z^2}$ of z_t^2 over the test days falls below one when the estimator overstates the risk.
- All tests use the t statistic t_{NW} with the variance estimator of Newey and West (1987).

Residual risk

Rule	Swaps	Equal	LW	OAS	EWMA 0.94	EWMA 0.98906
PCA	1	22.9	22.9	22.9	24.8	23.2
	2	2.56	2.56	2.56	2.32	2.34
Minimum variance	1	23.0	23.0	23.0	24.7	23.3
	2	2.16	2.93	2.69	2.18	2.09

Out-of-sample residual risk (%), median over 14 single tenors or 79 pairs.

- A single swap reduces the squared P&L by 75% to 77%, and two swaps eliminate 97% to 98%, regardless of the estimator.
- With one swap, both rules leave similar residual risk because PC1 accounts for most of the variance. No estimator outperforms the equal-weighted sample.
- Shrinkage yields the same PCA hedges as the equal-weighted estimator.

Hedge by hedge

			PCA		Minimum variance	
Swaps	Estimator	Hedges	Better	Worse	Better	Worse
1	LW	14	–	–	0	0
	OAS	14	–	–	0	0
	EWMA 0.94	14	0	4	0	4
	EWMA 0.98906	14	0	0	0	0
2	LW	79	–	–	0	38
	OAS	79	–	–	0	30
	EWMA 0.94	79	37	10	0	0
	EWMA 0.98906	79	36	6	1	0

Hedges with significantly lower (better) or higher (worse) squared P&L than under the equal-weighted estimator, Diebold and Mariano (1995) test at the 5% level. Under LW and OAS, PCA hedges equal those of the equal-weighted estimator and are not tested.

- Shrinkage makes the minimum-variance hedge with two swaps worse on 38 pairs under LW and 30 under OAS, and better on none.
- EWMA changes almost no minimum-variance hedge with two swaps, and is better on 36 or 37 PCA pairs and worse on 6 to 10.

Shrinkage and minimum-variance hedges

$$B^\top \hat{\Sigma} B = (1-\delta) B^\top S B + \delta\tau I, \qquad B^\top \hat{\Sigma} \boldsymbol{d} = (1-\delta) B^\top S \boldsymbol{d} + \delta\tau \boldsymbol{d}_B$$

- $\boldsymbol{d}_B = B^\top \boldsymbol{d}$ is the DV01 at the hedge tenors.
- The identity target alone would give $\boldsymbol{h}_{\mathrm{MV}} = -\boldsymbol{d}_B$, a hedge that offsets only the DV01 at the hedge tenors and ignores the correlation across the curve, and shrinkage pulls the hedge toward it.
- For two swaps whose rates move almost together, the smaller eigenvalue of $B^\top S B$ is small, so adding $\delta\tau$ to it changes the hedge materially.

Risk forecasts

	Unhedged		PCA, 2 swaps		Minimum variance, 2 swaps	
Estimator	$\overline{z^2}$	t_{NW}	$\overline{z^2}$	t_{NW}	$\overline{z^2}$	t_{NW}
Equal	0.732	−3.24	1.03	0.22	0.964	−0.30
LW	0.737	−3.17	0.237	−24.7	0.388	−9.05
OAS	0.735	−3.19	0.296	−18.4	0.408	−9.23
EWMA 0.94	0.961	−0.44	1.24	1.70	1.25	1.73
EWMA 0.98906	0.770	−2.83	0.984	−0.13	0.948	−0.44

Calibration of the risk forecast, with t_{NW} for H_0: $\mathbb{E}[z_t^2] = 1$. For hedged portfolios, z_t^2 is averaged over the 79 pairs on each day.

- Unhedged, the equal-weighted, LW and OAS forecasts overstate the risk because the rolling window still reflects the previous year while PC1 volatility decreases in 2025.
- EWMA with $\lambda = 0.94$ tracks the decline and shows no significant bias.
- Hedged, the equal-weighted forecast shows no significant bias, and shrinkage overstates the risk by a factor of about 2.5 to 4.

Why shrinkage inflates hedged risk

Proposition 1

Let $\hat{\Sigma} = (1-\delta)S + \delta\tau I$ with $0 \le \delta < 1$, and let S have distinct eigenvalues. Then (i) $\hat{\Sigma}$ and S have the same eigenvectors in the same order, so the PCA hedge is the same under both, and (ii) for every $\boldsymbol{q}$ with $\boldsymbol{q}^\top S\boldsymbol{q} > 0$,

$$\text{Inflation} = \frac{\boldsymbol{q}^\top \hat{\Sigma}\boldsymbol{q}}{\boldsymbol{q}^\top S\boldsymbol{q}} = (1-\delta) + \frac{\delta\tau}{\rho(\boldsymbol{q})}, \qquad \rho(\boldsymbol{q}) = \frac{\boldsymbol{q}^\top S\boldsymbol{q}}{\boldsymbol{q}^\top \boldsymbol{q}}$$

		Inflation	
Portfolio	$\rho(\boldsymbol{q})$ (bp^2)	LW	OAS
Unhedged	52.7	0.994	0.996
PCA, 2 swaps	0.108	4.47	3.43

Medians over the test days, and for PCA hedges also over the 79 pairs.

- An unhedged portfolio is dominated by the level factor and has a large $\rho(\boldsymbol{q})$, so the inflation is negligible.
- A hedged portfolio has its exposure to the leading components removed and a small $\rho(\boldsymbol{q})$, so $\delta\tau/\rho(\boldsymbol{q})$ inflates its forecast substantially.

Conclusion

- We evaluated whether covariance estimator selection affects the out-of-sample hedging and risk forecasting of a 14-tenor SOFR swap portfolio across 2025.
- No estimator offers a consistent hedging improvement over the equal-weighted sample. Shrinkage leaves PCA hedges unchanged and degrades minimum-variance hedges with two swaps, while EWMA improves 36 to 37 of the 79 PCA hedges with two swaps and worsens 6 to 10.
- Risk forecasting is sensitive to the estimator. Equal weighting overstates the unhedged risk as volatility falls, EWMA with $\lambda = 0.94$ shows no significant bias, and shrinkage overstates the residual risk of hedged portfolios by a factor of 2.5 to 4.
- The findings come from a single portfolio over 1 year, so the ranking of the estimators for the risk forecast may shift in high-volatility regimes.

References

- Chen, Y., Wiesel, A., Eldar, Y. C. and Hero, A. O. (2010). Shrinkage Algorithms for MMSE Covariance Estimation. *IEEE Transactions on Signal Processing*, 58(10), 5016–5029.
- Diebold, F. X. and Mariano, R. S. (1995). Comparing Predictive Accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263.
- J.P. Morgan/Reuters (1996). *RiskMetrics Technical Document*. 4th edition, New York.
- Ledoit, O. and Wolf, M. (2004). A Well-Conditioned Estimator for Large-Dimensional Covariance Matrices. *Journal of Multivariate Analysis*, 88(2), 365–411.
- Litterman, R. and Scheinkman, J. (1991). Common Factors Affecting Bond Returns. *Journal of Fixed Income*, 1(1), 54–61.
- Newey, W. K. and West, K. D. (1987). A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. *Econometrica*, 55(3), 703–708.

Code and replication data are available at *GitHub*.

Appendix A.1. Shrinkage intensities and test statistic

$$\delta_{\mathrm{LW}} = \frac{\min\left(\bar{b}^2, \|S - \tau I\|_F^2\right)}{\|S - \tau I\|_F^2}, \qquad \bar{b}^2 = \frac{1}{W^2}\sum_{s=1}^{W} \left\|\boldsymbol{x}_s \boldsymbol{x}_s^\top - S\right\|_F^2$$

$$\delta_{\mathrm{OAS}} = \min\left(1, \frac{\mathrm{tr}(S^2) + \mathrm{tr}^2(S)}{(W+1)\big(\mathrm{tr}(S^2) - \mathrm{tr}^2(S)/p\big)}\right)$$

- $\boldsymbol{x}_s$ are the demeaned changes in the estimation window and $\|A\|_F^2 = \mathrm{tr}(AA^\top)$.
- The condition number is $\kappa(A) = \sigma_{\max}(A)/\sigma_{\min}(A)$.

$$t_{\mathrm{NW}} = \frac{\bar{g}}{\sqrt{\hat{\Omega}/T}}, \qquad \hat{\Omega} = \hat{\gamma}_0 + 2\sum_{l=1}^{L}\left(1 - \frac{l}{L+1}\right)\hat{\gamma}_l, \qquad L = \left\lfloor 4(T/100)^{2/9} \right\rfloor$$

- $\hat{\gamma}_l$ is the sample autocovariance of the daily series g_t at lag l, with $g_t = \pi_{t,e}^2 - \pi_{t,\mathrm{Equal}}^2$ for the Diebold and Mariano (1995) test of estimator e and $g_t = z_t^2 - 1$ for the calibration test.